

Applied Statistics (MATH10096)

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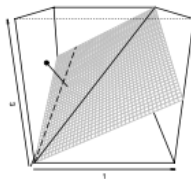
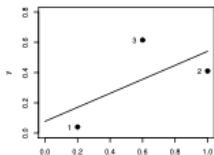


Semester 2, 2024/2025

Linear model theory (Section 4 of the lecture notes)

The geometry of linear models

- ↪ Least squares estimation of linear models amounts to finding the orthogonal projection of the n -dimensional response vector $\mathbf{y}$ onto the p -dimensional linear subspace spanned by the columns of $\mathbf{X}$.
- ↪ The linear model states that $\mathbb{E}(\mathbf{y}) = \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ lies in the space spanned by all possible linear combinations of the columns of the $n \times p$ model matrix $\mathbf{X}$, and least squares seeks the point in that space that is closest to $\mathbf{y}$ in Euclidean distance.



- ↪ **Left:** a straight line fit to three (x, y) data points.
- ↪ **Right:** the space in which the y coordinates of the data define a single point, while the columns of the model matrix ($p = 2$) (solid and dashed line) span the subspace shown in grey. The least squares estimate of $\mathbb{E}(\mathbf{y})$ is the orthogonal projection of the data point onto the model subspace.

Linear model theory (Section 4 of the lecture notes)

The geometry of linear models

↪ Given this view of least squares fitting as orthogonal projection it is interesting to look at the projection matrix that maps the data $\mathbf{y}$ to the fitted values $\hat{\boldsymbol{\mu}}$.

↪ We have

$$\hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{Q}\mathbf{R}\mathbf{R}^{-1}\mathbf{Q}^T\mathbf{y} = \mathbf{Q}\mathbf{Q}^T\mathbf{y}.$$

↪ So the required projection matrix is $\mathbf{A} = \mathbf{Q}\mathbf{Q}^T$ (*influence matrix* or *hat matrix* of the linear model).

↪ In terms of the design matrix $\mathbf{A} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$.

↪ One interesting property of $\mathbf{A}$ is that it is idempotent: $\mathbf{A}\mathbf{A} = \mathbf{A}$. Also, $\mathbf{A} = \mathbf{A}^T$.

Linear models in R

The `cars` dataset

- ↪ Let us consider the `cars` data frame, which contains **data on stopping distances of cars against speed when the driver was first signalled to stop**.
- ↪ It typically takes some fixed amount of 'reaction time', after the stop signal, for the driver to apply the brakes, hence the car will travel a distance proportional to initial speed before beginning to slow at all: so we need a linear dependence on `speed` in the model.
- ↪ The kinetic energy of a car is proportional to the square of its `speed`, and once the brakes are applied they can dissipate this energy (ultimately as heat) at a constant rate per unit distance travelled: hence the braking phase contributes a distance proportional to the square of `speed` to the total stopping distance.
- ↪ A suitable model for the data is then:

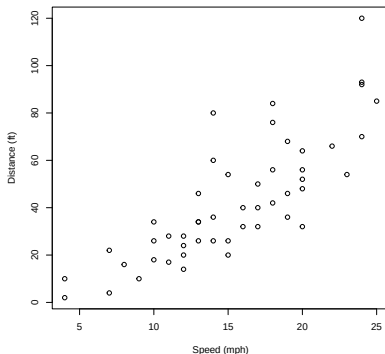
$$\text{distance}_i = \beta_1 + \beta_2 \text{speed}_i + \beta_3 \text{speed}_i^2 + \epsilon_i, \quad \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

- ↪ Of course we would expect the parameter β_1 to be zero, given the justification for the model, but there is no harm in leaving it in for the moment.

Linear models in general (Section 3 of the lecture notes)

R and linear models in general: a quadratic model

```
data(cars)
#Scatter plot of the data
plot(cars$speed, cars$dist,
      xlab = "Speed (mph)", ylab = "Distance (ft)")
```



Linear models in general (Section 3 of the lecture notes)

R and linear models in general: a quadratic model

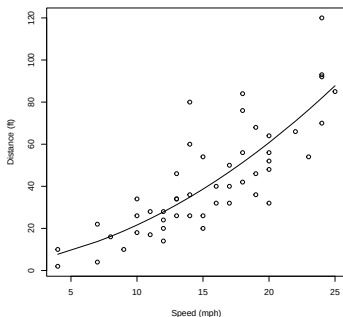
↪ We start by fitting the quadratic regression model

```
data(cars)
cm <- lm(dist ~ speed + I(speed^2), data = cars)
```

Linear models in general (Section 3 of the lecture notes)

R and linear models in general: a quadratic model

```
plot(cars$speed, cars$dist,  
      xlab = "Speed (mph)", ylab = "Distance (ft)")  
  
ss <- sort(cars$speed)  
lines(ss, coef(cm)[1] + coef(cm)[2]*ss + coef(cm)[3]*ss^2)
```



Linear models in R

Confidence intervals and hypothesis tests for single parameters

- ↪ We learned about the following result when we have p terms in a linear model (and n observations):

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}_{\hat{\beta}_j}} \sim t_{n-p}.$$

- ↪ We will now use this result to find confidence intervals for, and test hypotheses about, model parameters.

Linear models in R

Confidence intervals and hypothesis tests for single parameters

- From the resulting fitted model object, `cm`, it is possible to extract all the quantities needed to construct the T statistics and to compute confidence intervals.

```
beta.hat <- coef(cm)
beta.hat

## (Intercept)      speed  I(speed^2)
##  2.4701378    0.9132876    0.0999593

V.beta <- vcov(cm)
V.beta

##           (Intercept)      speed  I(speed^2)
## (Intercept) 219.5483705 -28.9523122  0.872858710
## speed       -28.9523122   4.1380528 -0.131439753
## I(speed^2)   0.8728587  -0.1314398  0.004351805

sigma.beta <- sqrt(diag(V.beta))
sigma.beta

## (Intercept)      speed  I(speed^2)
## 14.81716473   2.03422044  0.06596821
```

Linear models in R

Confidence intervals

- Let's start by finding a 90% CI for β_2 .
- This will involve using the `qt` function to evaluate the value above which 5% of t_{47} (note that $n = 50$ and $p = 3$) distributed random variables lie (a further 5% lying below the negative of that value, by symmetry of the t distribution).

```
df <- nrow(cars) - 3
qt(0.95, df = df)
## [1] 1.677927
```

- Thus the 90% CI for β_2 is

$$\hat{\beta}_2 - 1.678\hat{\sigma}_{\hat{\beta}_2} \leq \beta_2 \leq \hat{\beta}_2 + 1.678\hat{\sigma}_{\hat{\beta}_2}.$$

```
beta.hat[2] + qt(c(0.05, 0.95), df = df)*sigma.beta[2]
## [1] -2.499985 4.326560
```

Linear models in R

Confidence intervals and hypothesis tests for single parameters

- Actually, it is possible to calculate intervals for all three parameters at the same time.
- The following finds 95% CI's for all the parameters.

```
beta.hat - qt(0.975, df = df)*sigma.beta
##      (Intercept)      speed      I(speed^2)
## -27.33815279    -3.17903606    -0.03275162
beta.hat + qt(0.975, df = df)*sigma.beta
##      (Intercept)      speed      I(speed^2)
##   32.2784284      5.0056113      0.2326702
```

- So, for example, a 95% CI for β_3 is $(-0.033, 0.233)$.

Linear models in R

Confidence intervals and hypothesis tests for single parameters

↪ Note that the `confint` function calculates these confidence intervals automatically.

```
confint(cm, level = 0.95)
##              2.5 %      97.5 %
## (Intercept) -27.33815279 32.2784284
## speed       -3.17903606  5.0056113
## I(speed^2)   -0.03275162  0.2326702
# Also, for 90% CI for beta2
confint(cm, level = 0.9)[2, ]
##          5 %      95 %
## -2.499985  4.326560
```

Linear models in R

Confidence intervals and hypothesis tests for single parameters

↪ The result derived is also useful for testing hypotheses about the values of the β_j , $j = 1, 2, 3$.

↪ Let us start by testing H_0 :

$$\beta_2 = 0 \quad \text{vs.} \quad H_1 : \beta_2 \neq 0.$$

↪ To test this we first form the t-statistic, $T = \hat{\beta}_2 / \hat{\sigma}_{\hat{\beta}_2}$, and compute its observed value.

```
obs_t <- beta.hat[2]/sigma.beta[2]
obs_t
##      speed
## 0.448962
```

↪ If H_0 is true then this should be an observation of a t_{47} random variable,

↪ But if H_1 is true then it should have too large a magnitude for this (basically because $\hat{\beta}_2$ will not then be 'close' to the hypothesized value, zero).

Linear models in R

Confidence intervals and hypothesis tests for single parameters

- As usual we quantify the evidence for or against H_0 using a p-value.
- This is given, in this case, by the following probability: $\Pr[|T| > |t|]$, where $T \sim t_{47}$ and $t = 0.449$.

```
p.val <- 2*pt(abs(obs_t), df = df, lower.tail = FALSE)
p.val
##      speed
## 0.6555224
```

- A p-value of 0.66 suggests that the data are quite compatible with H_0 .
- That is, provided that the model contains β_1 and β_3 terms, β_2 seems to be superfluous.
- Note that the 95% CI for β_2 , $(-3.179, 5.006)$, contains zero. This implies that we would not reject the null hypothesis that $\beta_2 = 0$ at the 5% significance level.

Linear models in R

Confidence intervals and hypothesis tests for single parameters

- ↪ The previous test did not provide evidence to reject $H_0 : \beta_2 = 0$.
- ↪ However it might be that one of β_1 and β_3 is even more superfluous, so we should test whether either of those might be zero, before dropping any terms.
- ↪ In fact we can work out equivalent p-values for all 3 parameters simultaneously.

```
obs_t <- beta.hat/sigma.beta
p.val <- 2*pt(abs(obs_t), df = df, lower.tail=F)
p.val
```

## (Intercept)	speed	I(speed^2)
## 0.8683151	0.6555224	0.1364024

Linear models in R

Confidence intervals and hypothesis tests for single parameters

→ Actually, p-values for testing whether parameters might be zero are very often useful, so R produces them by default in the `summary` of a fitted linear model.

```
summary(cm)$coefficients
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	2.4701378	14.81716473	0.1667079	0.8683151
## speed	0.9132876	2.03422044	0.4489620	0.6555224
## I(speed^2)	0.0999593	0.06596821	1.5152647	0.1364024

Linear models in R

Confidence intervals and hypothesis tests for single parameters

- ↪ These single parameter tests are often used to decide whether to drop a term from the model: usually the term with the highest p-value greater than some threshold, such as 0.05, would be dropped (provided that there are no prior grounds for insisting on its retention in the model).
- ↪ The model is refitted without the term, and the new p-values re-examined to see if any further simplification is possible.
- ↪ This is repeated until all terms have p-values lower than, say, 0.05.
- ↪ Note that the different estimates in $\hat{\beta}$ are not independent (see the covariance matrix in a previous slide) and then dropping one term (setting it to zero) will change the estimates of the other coefficients and hence their p -values.

Linear models in R

Confidence intervals and hypothesis tests for single parameters

- Sometimes other types of hypotheses may be tested. For example, it is hard to think of a decent physical interpretation for a negative β_1 .
- Hence it might be more appropriate to test $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 > 0$.

```
obs_t <- beta.hat[1]/sigma.beta[1]
obs_t
## (Intercept)
## 0.1667079
```

- In this case large magnitude negative values of T count for H_0 , so only the upper tail of the t_{47} distribution features in the p-value calculation: $\Pr[T > t]$ where $T \sim t_{47}$ and $t = 0.167$.

```
p.val <- pt(obs_t, df = df, lower.tail = F)
p.val
## (Intercept)
## 0.4341575
```

- Again, no reason to reject the null (provided the other terms are in the model), but notice that the p-value is halved by the change in alternative.

Linear models in R

Confidence intervals and hypothesis tests for single parameters

- ↪ Another use of the result derived is to test whether $\beta_j, j = 1, 2, 3$, might have some hypothesized value other than zero.
- ↪ For example we might be able to calculate what β_3 ought to be, based on the frictional properties of the brakes and tyres: suppose this calculation says that $\beta_3 = 0.2$.
- ↪ We can test this theoretical value, by testing $H_0 : \beta_3 = 0.2$ vs. $H_1 : \beta_3 \neq 0.2$.

```
obs_t <- (beta.hat[3] - 0.2)/sigma.beta[3]
p.val <- 2*pt(abs(obs_t), df = df, lower.tail = F)
p.val
## I(speed^2)
## 0.136091
```

- ↪ Not much evidence with which to reject H_0 .

Linear models in R

Confidence intervals and hypothesis tests for single parameters

- This test provides a very good example of why it is often a good idea to base statistical conclusions on the simplest model consistent with the data.
- If we repeat the test, based on a model without an intercept (which based on physical grounds should be zero), then things are rather different.

```
cm1 <- lm(dist ~ speed + I(speed^2) - 1, data = cars)
beta.hat <- coef(cm1)
V.beta <- vcov(cm1)
sigma.beta <- sqrt(diag(V.beta))
obs_t <- (beta.hat[2] - 0.2) / sigma.beta[2]
p.val <- 2 * pt(abs(obs_t), df = (nrow(cars) - 2), lower.tail = F)
p.val

##      I(speed^2)
## 0.0004934814
```

- The p-value has dropped enormously, and we now have firm evidence to reject $H_0 : \beta_3 = 0.2$.

Linear models in R

F-ratio tests about several parameters

- ↪ We will keep considering, for the sake of illustration, the quadratic model for the `cars` dataset:

$$\text{dist}_i = \beta_1 + \beta_2 \text{speed}_i + \beta_3 \text{speed}_i^2 + \epsilon_i, \quad \epsilon_i \text{ i.i.d. } N(0, \sigma^2).$$

- ↪ Suppose that we had some prior reason to want to test:

$$H_0 : \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{vs} \quad H_1 : \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

- ↪ This is exactly the situation in which the results of section 4.3.2 are useful. Recall that the result is:

$$F = \frac{(\text{RSS}_0 - \text{RSS}_1)/(p_1 - p_0)}{\hat{\sigma}^2} \sim F_{p_1 - p_0, n - p_1} \text{ if } H_0 \text{ is true.}$$

- ↪ Here $p_1 - p_0$ is the difference in degrees of freedom between the models (difference in number of parameters) and $\hat{\sigma}^2 = \text{RSS}_1/(n - p_1)$ is the residual variance estimator from the alternative (larger) model.

Linear models in R

F-ratio tests about several parameters

→ Fitting the null model, it is then straightforward to calculate the F ratio statistic:

```
cm <- lm(dist ~ speed + I(speed^2), data = cars)
cm0 <- lm(dist ~ I(speed^2) - 1, data = cars)
rss0 <- sum(residuals(cm0)^2)
rss1 <- sum(residuals(cm)^2)
f_stat <- (rss0 - rss1)/2)/(rss1/47)
f_stat
## [1] 2.412267
```

→ From the result derived before, this ought to be an observation on an $F_{2,47}$ random variable, if H_0 is true.

→ As usual we quantify the compatibility of the test statistic (and hence the data) with the stated distribution by calculating a p-value, which in this case is given by $\Pr[F > f]$ where $F \sim F_{2,47}$ and $f = 2.412$.

```
p.val <- pf(f_stat, 2, 47, lower.tail = F)
p.val
## [1] 0.1006278
```

Linear models in R

F-ratio tests about several parameters

↪ The test just conducted can be done using the `anova` function as follows.

```
anova(cm0, cm)

## Analysis of Variance Table
##
## Model 1: dist ~ I(speed^2) - 1
## Model 2: dist ~ speed + I(speed^2)
##   Res.Df  RSS Df Sum of Sq    F Pr(>F)
## 1      49 11936
## 2      47 10825  2    1111.2 2.4123 0.1006
```

Linear models in R

F-ratio tests about several parameters

- ↪ Similar conclusions are obtained if we revert the order of the two models in the `anova` command.

```
anova(cm, cm0)

## Analysis of Variance Table
##
## Model 1: dist ~ speed + I(speed^2)
## Model 2: dist ~ I(speed^2) - 1
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1       47 10825
## 2       49 11936 -2   -1111.2  2.4123 0.1006
```


Linear models in R

F-ratio tests about several parameters

↪ Note that, as claimed in Section 4.3.2, the F test statistic in the way we have just computed it is equivalent to

$$F = \frac{1}{q}(\mathbf{C}\hat{\beta} - \mathbf{d})^T(\mathbf{C}\hat{V}_{\hat{\beta}}\mathbf{C}^T)^{-1}(\mathbf{C}\hat{\beta} - \mathbf{d}),$$

where in this case $q = 2$ and

$$\mathbf{C} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \text{and} \quad \mathbf{d} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

```
beta.hat <- coef(cm)
V.beta <- vcov(cm)
C <- matrix(c(1, 0, 0, 0, 1, 0), nrow = 2, ncol = 3, byrow = T)
d <- matrix(c(0, 0), ncol = 1, nrow = 2)
q <- 2
as.numeric(((t(C%*%beta.hat - d))%*%solve(C%*%V.beta%*%t(C))%*%
           (C%*%beta.hat - d))/q)
## [1] 2.412267
```

F-ratio tests about several parameters

F-ratio tests about several parameters

- It is worth mentioning that the last line in the `summary` object gives an F -statistic where the null hypothesis is that all coefficients, except the intercept, are zero.
- In this case this corresponds to $H_0 : \beta_2 = \beta_3 = 0$.

```
summary(cm)
##
## Call:
## lm(formula = dist ~ speed + I(speed^2), data = cars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -28.720  -9.184  -3.188   4.628  45.152
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.47014    14.81716   0.167   0.868
## speed        0.91329     2.03422   0.449   0.656
## I(speed^2)   0.09996     0.06597   1.515   0.136
##
## Residual standard error: 15.18 on 47 degrees of freedom
## Multiple R-squared:  0.6673, Adjusted R-squared:  0.6532
## F-statistic: 47.14 on 2 and 47 DF,  p-value: 5.852e-12
```

Linear models in R

F-ratio tests about several parameters

→ The following is possible.

```
anova(cm)

## Analysis of Variance Table
##
## Response: dist
##           Df Sum Sq Mean Sq F value    Pr(>F)
## speed      1 21185.5  21185.5   91.986 1.211e-12 ***
## I(speed^2)  1   528.8    528.8    2.296   0.1364
## Residuals 47 10824.7    230.3
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

→ The tables have to be read from bottom row up.

→ This single argument call causes `anova` to successively remove terms from your model, at each stage testing the null hypothesis that the current version of the model is correct, against the alternative that the previous version was right.

→ Actually this single argument use of `anova` only really makes sense in very specific situations. As we have already seen, if we reduce the `cars` model in a sensible order, we would drop the intercept term, and then nothing else.